

Airpay Academy — Production Database Diagnostic

Analysis of airpayprod backup (6th April 2026)

Prepared by: L&D Technology Team | **Date:** April 2026 **Source:** airpayprod-6-4.sql (3.5GB MySQL dump) **DB Host:** lms-prod-db.crpst4qn6rtu.ap-south-1.rds.amazonaws.com (MySQL 8.0.44, AWS RDS ap-south-1)

Executive Summary

This report analyzes the production database of Airpay Academy (www.airpay.academy) to identify redundancies, structural limitations, and operational pain points across the multi-tenant LMS deployment. The analysis covers 3,374 users, 443 courses, 881 SCORM packages, and 3 organizational tenants.

Top 5 systemic issues identified:

#	Issue	Impact
1	No cross-tenant content deployment	72 courses manually duplicated; every update doubles work
2	SCORM storage bloat	3.3x duplication; 881 packages from 263 unique files
3	Learning plan sync burden	Same plans maintained separately per tenant
4	Orphaned users	154 users (8.5%) on non-existent organizational paths
5	Flat permission model	1 admin, 4,106 employees, no middle management layer

Platform Overview

Metric	Value	Notes
Tables	580	Full Moodle 4.x + BizLMS
Users	~3,374	Active across 3 tenants
Courses	~443	Distributed across tenants
User Enrolments	~52,000	Across all courses
Course Completions	~56,000	Historical records
SCORM Packages	~881	263 unique, rest duplicated
SCORM SCOs	~2,000	Learning objects
Costcenter Records	213	3 tenants, 4 levels deep
Learning Plans	20+	Spanning multiple tenants
Certificates Issued	8,822	Course + learning plan types
Quiz Attempts	~308,000	Across 260 quizzes
Question Categories	954	Supporting 260 quizzes

Tenant / Org Structure

ID	Name	Shortname	Path	Sub-Orgs	Users (est.)	Courses
1	AIRPAY PAYMENT SERVICES PVT LTD	AirPay	/1	205	~2,200	188
77	Public	external	/77	1	~760	184
177	ZEEA	ZEEA01	/177	4	~15	15
-	Orphaned (/80, /100)	-	-	-	154	1

Airpay Department Hierarchy (205 sub-organizations)

- /1 AIRPAY PAYMENT SERVICES PRIVATE LIMITED
 - /1/2 Airpay Payment Services (Airpay Acquiring)
 - /1/2/3 Acquiring Sales (Key Accounts, New Business, Sales-Govt)
 - /1/2/4 Alliances and Partnerships
 - /1/2/5 Emerging Business (International Sales)
 - /1/2/6 Finance (AP, AR, Audit, Taxation)
 - /1/2/7 Human Resources (Admin, Culture, Operations, L&D, Payroll)
 - /1/2/8 HOD (Management)
 - /1/2/9 Marketing
 - /1/2/10 Operations

- /1/2/11 Product
- /1/2/12 Technology (sub-departments at depth 4)
- /1/79 Vyaapaar Fintech (774 users on /1/79/93/97 alone)
- /1/114 AFT and other subsidiaries

DIAGNOSTIC 1: Content Duplication Tax

The "Build Once, Upload Twice" Problem

Metric	Value
Courses on Airpay tenant (/1)	188
Courses on Public tenant (/77)	184
Courses duplicated across BOTH	72 (with different shortnames)
Courses with SAME shortname across tenants	0
SCORM packages (total)	881
Unique SCORM zip files	263
SCORM duplication factor	3.3x
Most duplicated SCORM file	story.zip - 74 instances

Examples of Duplicated Courses

Course Title	Airpay Shortname	Public Shortname	Copies
Anti Money Laundering	AMLE001	AMLI01	3
Critical Thinking	CTE001	CTE0001 / CTI002	3
Communication Skills	EESCS0001	EESCS001	2
Financial Numeracy Skills	(variant)	(variant)	3
HR Onboarding	A0002	A0009	2
Time Management	TM0001	TM0001E	2

Key finding: Zero courses share the same shortname across tenants. This confirms 100% manual re-creation with no automated cross-tenant deployment.

Questions for Discussion

1. What is the current workflow for deploying a course to both Airpay and Public tenants?
2. How many hours per month does L&D spend re-uploading the same content to the second tenant? At 72 duplicated courses, even 30 min each = 36 hours historically.
3. When a SCORM package is updated (e.g., AML compliance changes), how do you ensure both tenant copies get updated?
4. Are there courses that SHOULD be cross-tenant but are not yet because of the manual effort?
5. The 112 Public-only and 146 Airpay-only courses: are those genuinely tenant-specific, or a backlog?

DIAGNOSTIC 2: Learning Plan Mirroring

Same Plan, Two Tenants

Learning Plan	Airpay Copy	Public Copy	Course Count
Customer Success Foundation	LP06 (/1), 28 completions	LP02 (/77), 11 completions	37 vs 42 (diverged)
HR Onboarding	HOVC01 (/1, "Vyaapaar"), 38 courses	HROBC01 (/77), 38 courses	Same count
Employability Skills	-	LEE001 (/77), 42 courses	Public only
Product Training Tanzania	PTT001 (/1), 24 courses	-	Airpay only
ERP Products	VP01 (/1), 24 courses	-	Airpay only
PG Products and Features	PGP01 (/1), 24 courses	-	Airpay only
Business Correspondents	-	BCBF01 (/77), 48 courses	Public only

Questions for Discussion

6. Customer Success Foundation Path exists in both tenants with DIFFERENT course counts (37 vs 42). Why the divergence?
7. HR Onboarding has two copies with identical course counts but different shortnames. What is the maintenance burden when curriculum changes?
8. Who is accountable for keeping mirrored learning plans in sync?
9. If you could deploy one learning plan to multiple tenants simultaneously, which plans would benefit most?

DIAGNOSTIC 3: SCORM Content Sprawl

Pattern	Count
Unique SCORM zip files	263
Total SCORM instances	881
story.zip instances	74
Banking/financial series (5+ copies each)	~15 modules
Course 49 alone	351 SCORM packages
Course 58 alone	117 SCORM packages

Most Duplicated SCORM Packages

- 4.12_Red Flags_Ver01.zip - 6 instances
- 1.9_Role of Banks in financial inclusion_Ver01.zip - 5 instances
- 1.8_Recent Trends In Banking_Ver01.zip - 5 instances
- 1.7_Evolution of Banking in India_Ver01.zip - 5 instances
- Plus ~10 more banking modules with 5 copies each

Questions for Discussion

10. Course 49 has 351 SCORM packages. Is this a content library/assessment hub, or accidental accumulation?
11. The banking/financial compliance series appears 5-6 times. Same content in different courses, or tenant-specific variants?
12. story.zip appears 74 times. Is this an Articulate Storyline default export name that should have been renamed?
13. When RBI issues an AML regulation update, how many SCORM packages need individual updating?

DIAGNOSTIC 4: Assessment Reuse Opportunity

External Hiring Assessments vs Internal Skill Assessments

Metric	Value
Total quizzes	260
Courses with quizzes	64
Question categories	954
Question bank attempts	~308,000
Attempt step data	~1.1 million records

Questions for Discussion

- 14. The Public tenant (hiring/external) has assessment courses. Are any measuring the same competencies as internal Airpay skill assessments?
- 15. 954 question categories for 260 quizzes: is there a taxonomy or naming convention? A structured question bank could serve both hiring and internal assessment.
- 16. Are hiring assessments ever shared back into internal learning paths as pre/post tests?
- 17. With 308K quiz attempts: which assessments have the highest failure rates?

DIAGNOSTIC 5: Orphaned Data and Integrity Gaps

Issue	Count	Details
Users on path /100 (no costcenter exists)	113	Not in mdl_local_costcenter
Users on paths /80, /81 (no costcenter exists)	41	Not in mdl_local_costcenter
Total orphaned users	154	8.5% of active users
Orphaned course on path /80	1	Also has no matching costcenter
User info custom fields	0 records	mdl_user_info_field table is empty
Costcenter color fields	All NULL	brand_color, button_color, hover_color unused

User Distribution by Path

Path	Users	Percentage	Organization
/1/79/93/97	774	42.5%	Vyaapaar Fintech
/77	423	23.2%	Public/External
/100 (orphaned)	113	6.2%	Unknown - no costcenter
/1 (root)	42	2.3%	Airpay root
/1/114/122/*	~100	5.5%	AFT subsidiaries
/80-81 (orphaned)	41	2.3%	Unknown - no costcenter

Questions for Discussion

- Who are the 154 users on paths /100 and /80-81? Former tenants deleted? Migration artifacts? Test accounts?
- Is there a user lifecycle process when an employee moves between subsidiaries?
- The costcenter table has brand_color, button_color, hover_color fields, all NULL. Was per-tenant branding planned but never implemented?

DIAGNOSTIC 6: Tenant Architecture Limitations

Structural Observations

- BizLMS enforces open_path scoping: a course on /1 is invisible to /77 users
- No "shared" or "global" tenant concept exists. Courses MUST belong to one path
- Enrolment is standardized: manual + self + guest + learning plan on every course
- 99.98% of role assignments are "Employee" (role 5): very flat permission model
- Only 1 Administrator role assignment in the entire system (out of 4,107 total)

Questions for Discussion

21. If you could create a virtual "shared" tenant whose courses are visible to ALL tenants, which courses would go there?
22. With only 1 admin role and 4,106 Employee roles: who manages content for each tenant? Are there L&D coordinators per subsidiary who lack admin access?
23. ZEEA (id=177) has 15 courses and ~15 users. Is this ramping up, a pilot, or dormant?
24. Vyaapaar Fintech has 774 users on one leaf path. Should it have departmental structure like Airpay Acquiring?
25. The Public tenant shortname is "external". Does it serve ONLY hiring candidates, or also partners, vendors, BCs?

DIAGNOSTIC 7: Certificate and Completion Maturity

Metric	Value
Certificates issued	8,822
Certificate templates	5+
Certificate types	course and learningplan
Course completions	~56,000 records
Top course completions	Course 2058: 92, Course 935: 55

Questions for Discussion

- 26. 56K completions across 443 courses = 126 avg, but distribution is extremely skewed. What percentage of courses have ZERO completions?
- 27. 8,822 certificates from 3,374 users = 2.6 per user average. Is this meeting the L&D target?
- 28. Are certificates used for compliance evidence (e.g., RBI audit trail)?

DIAGNOSTIC 8: Operational Workflow Friction

Questions for Discussion

- 29. What is the current course creation workflow end-to-end? How many steps are manual?
- 30. When onboarding a new subsidiary (like ZEEA), what is the process and how long does it take?
- 31. Is there a content review/approval workflow before courses go live?
- 32. How are course updates communicated to employees?
- 33. Is there cross-tenant reporting today? Can you see "total AML completion rate across ALL subsidiaries" in one view?

Summary: Top 5 Systemic Issues

#	Issue	Impact	Data Evidence
1	No cross-tenant content deployment	72 courses manually duplicated; every update doubles	0 shared shortnames; different codes for same content
2	SCORM storage bloat	3.3x duplication; 881 packages from 263 unique files	story.zip x74; banking series x5-6 each
3	Learning plan sync burden	Same plans maintained separately per tenant	Customer Success: 37 vs 42 courses; HR Onboarding: 2 copies
4	Orphaned users	154 users (8.5%) on non-existent paths	/100 (113 users), /80-81 (41 users) with no matching costcenter
5	Flat permission model	1 admin, 4,106 employees, no middle tier	No L&D coordinators, no content managers per subsidiary

Roadmap Implications

These diagnostics inform three layers of solution:

- 1. Theme deployment (Phase 15, immediate):** airpayux must handle ZEEA, orphaned paths gracefully, and the flat role model
- 2. BizLMS enhancement (Phase 8+):** Cross-tenant course deployment, shared content tier, SCORM deduplication
- 3. SENTIENTIA pipeline (Workstream B):** Build once, auto-deploy to selected tenants, eliminating the manual duplication tax

Technical Reference

Production Config vs Local Dev

Setting	Production	Local Dev	Action Needed
theme	epsilon	airpayux	Change on deploy
defaulthomepage	1 (Dashboard)	0 (Site home)	Change
authloginviaemail	0	1	Change
enablewebservices	1	1	Already enabled

Critical Schema Finding

The `open_costcenterid` column does NOT exist on production user or course tables. BizLMS resolves tenant membership dynamically at runtime from `open_path` lookup against `mdl_local_costcenter`. No database schema changes are required for theme deployment.

This analysis was generated from the production database backup dated 6th April 2026. All data is read-only; no modifications were made to the production system.